NBA Game Outcome Predictor

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https://github.com/neonpancake624/NBA_game_predictor

- Project goal

The goal of this project is to build a ML app that predicts the outcome of NBA games using historical and current team performance data. In addition to predicting which team is more likely to win, the app aims to provide insights for why a certain team is favored to win a matchup. These explanations are based on performance metrics such as shooting efficiency, rebounding advantage, assists, turnovers, point differential, and home court advantage.

- My approach

My approach is data cleaning, feature engineering, EDA, model training, model evaluation, prediction, and reasoning.

  - Data cleaning:
    - Removing missing values
    - Target variable home_win for whether or not the home team won or lost
  - Feature engineering
    - Calculated statistical differences between the home and away teams
      - Free throw, assist, turnover, field goal
  - EDA
    - Created visuals to analyze data
      - Home/away wins
      - Score difference distributions
      - Correlations between statistical features and game outcomes
  - ML model
    - logistic regression for the primary classification model
  - Prediction w/explanation
    - Interpretable explanation for why a team was favored
      - Shooting stats
      - Rebounding advantage
      - Assist
      - Home court
  - Interactive matchup choice
    - Users can select two teams and generate a winner, view the win probability, and get an explanation for why a team was favored

- Finished features
  - Data loading and preprocessing
  - EDA
  - Logistic regression classification model

    - Win probability prediction and prediction explanation system
    - Streamlit web app

- Evaluations/results
    - The logistic regression model was evaluated using accuracy and confusion matrix. The original model used same game statistics, which isn't correct, and produced unrealistic predictions and metrics. The current model, using team average statistics, produced more realistic predictions.
- Limitations
    - Right now, I'm not using player statistics, which could improve prediction accuracy and give more accurate predictions. I also don't account for back to back games, which are very frequent and can cause teams to lose games because of fatigue.